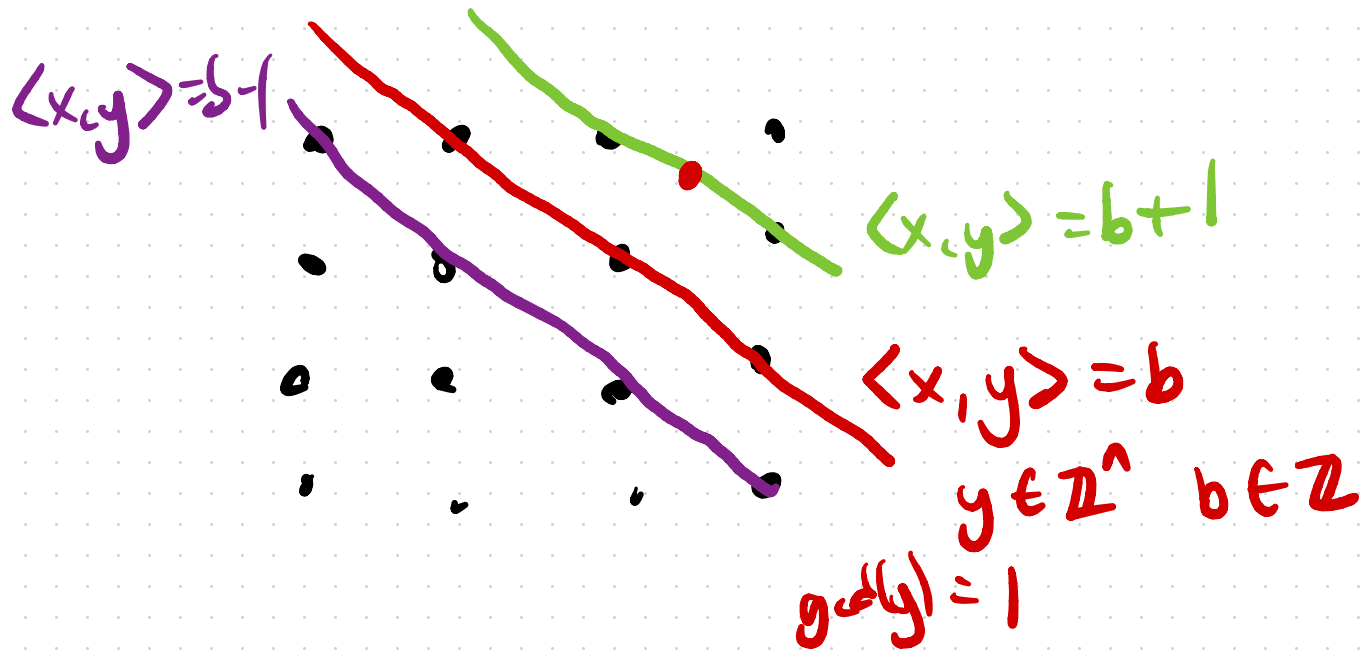
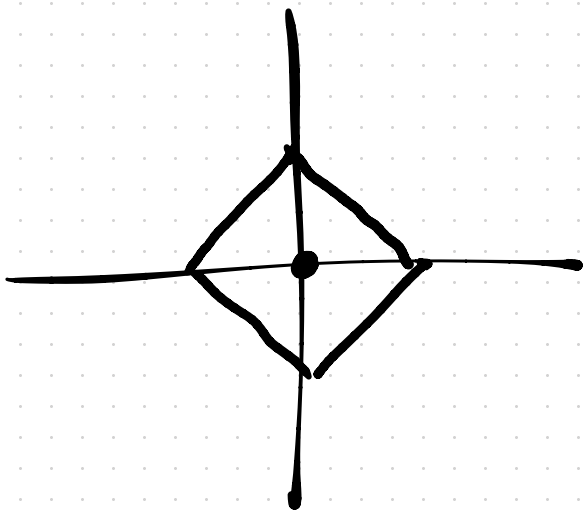


An integral polytope is **reflexive** if it contains a unique lattice point in its relative interior, and this point is **lattice distance** 1 from all its facets.





$$P = \{ \langle x, y_i \rangle \leq 1 \text{ for all } i \}$$

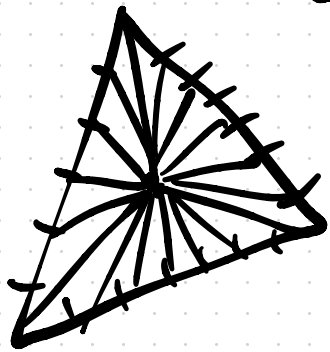
↑

$$y_i \in \mathbb{Z}^n$$

$$\gcd(y_i) = 1$$

Thm: $h_i^* = h_{d-i}^*$ iff P is reflexive and dimension d .

Note: If $2P$ is reflexive and has a unimodular triangulation, then coning over this triangulation with cone point the origin gives



P

a unimod. triangulation of P whose h -vector is the same as the h -vector of the boundary triangulation, and therefore symmetric. This is the same as $h^*(P)$.

Proof:

$$H(A(P), t) = \frac{h_0^* + h_1^* t + \dots + h_d^* t^d}{(1-t)^{d+1}}$$

$$H(A(P^0), t) = \frac{h_d^* t + h_{d-1}^* t^2 + \dots + h_0^* t^{d+1}}{(1-t)^{d+1}}$$

So $h_i^* = h_{d-i}^*$ for all i iff

$$tH(A(P), t) = H(A(P^0), t)$$

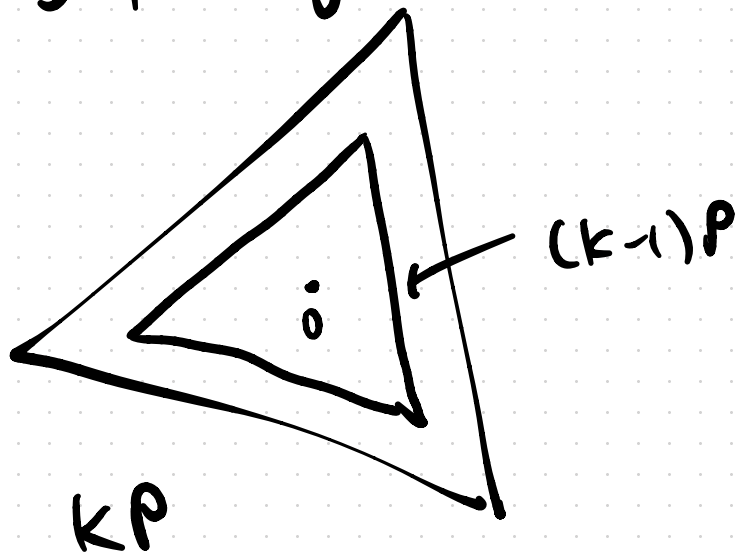
The coefficient of t^k in $H(A(P), t)$ is the
of lattice points in kP .

$$\therefore \quad t^k \text{ in } H(A(P^0), t) \quad ,$$
$$kP^0$$

So the above occurs iff for all $k \geq 1$

$$\#((k-1)P \cap \mathbb{Z}^n) = \#(kP^0 \cap \mathbb{Z}^n)$$

For $k=1$, this says that P has a unique lattice point in its interior. WLOG, assume this point is the origin.



Thus $(k-1)P \cap \mathbb{Z}^n = KP \cap \mathbb{Z}^n$ for all k , so

$$(kP \setminus (k-1)P) \cap \mathbb{Z}^n = \emptyset.$$

For large enough k , this can only happen if the corresponding facets of kP and $(k-1)P$ are lattice distance 1 apart.

So each facet of P must be ^{lattice} distance 1 from the origin.

Then: There are integral polytopes with non-unimodal h^* -vectors.

Then: There are reflexive polytopes with non-unimodal h^* -vectors.

A ^{integral} polytope P has the integer decomposition property if $A(P)$ is generated in degree 1. (IDP)

I.e., every lattice point in kP , $k \in \mathbb{Z}_{>0}$, can be written as a sum of k elements of $P \cap \mathbb{Z}^n$.

Example: Unimodular simplex, parallelepipeds, anything that can be written as the union of smaller IDP polytopes

Conjecture: If P has IDP, then $h^*(P)$ is unimodal.

Unpublished proof for reflexive IDP (Adiprasito -
Papadakis - Petrotou 2025).

Open even assuming P has a unimodular triangulation.